

# WEEK 03 INTERNSHIP REPORT

## Full Stack Development Internship

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|-----------------|--------------------------------------------------------------------------|
| Intern          | Arsh Kamal                                                               |
| Organization    | UpSkill Campus × UniConverge Technologies (UCT)                          |
| Week            | Week 03                                                                  |
| Technical Topic | Java Database Connectivity (JDBC)                                        |
| Learning Focus  | JDBC Statements, limitations of Simple Statement, and Prepared Statement |

### 1. Overview

During Week 03, the learning session focused on Java Database Connectivity (JDBC). The supplied training material covered the JDBC learning roadmap, the three JDBC statement objects, limitations of the Simple Statement, and the concept and working of the Prepared Statement.

### 2. Topics Covered

- JDBC and its role in connecting Java applications with databases.
- Three types of JDBC statement objects: Statement, PreparedStatement, and CallableStatement.
- Limitations of the Simple Statement when queries contain variables and when the same query is executed repeatedly.
- Prepared Statement as an approach based on pre-compiled SQL queries.
- Repeated execution of a prepared query with the same or different values.

### 3. Detailed Learning

#### Types of JDBC Statements

The training material identifies three JDBC statement objects: **Simple Statement Object (Statement)**, **Prepared Statement Object (PreparedStatement)**, and **Callable Statement Object (CallableStatement)**.

#### Limitations of Simple Statement

According to the session material, framing an SQL query with variables using a Simple Statement can be complex. When the same query is executed multiple times, the query is sent to the database software multiple times, parsed multiple times, executed multiple times, and its output is fetched multiple times.

#### Prepared Statement

PreparedStatement deals with pre-compiled SQL queries. The material explains that the query reaches the database software only once and is parsed only once. The prepared query can then be executed

multiple times with the same or different values and results can be gathered multiple times. This process is described in the material as a pre-compiled query.

## **4. Activities Completed**

- Studied the JDBC learning roadmap.
- Identified and reviewed the three JDBC statement types.
- Studied the limitations associated with Simple Statement.
- Learned the purpose and basic working concept of PreparedStatement.
- Compared repeated query processing in Simple Statement with the pre-compilation approach of PreparedStatement.

## **5. Key Learning Outcomes**

By the end of the session, I developed a clearer understanding of JDBC statement objects and why PreparedStatement is used for handling SQL queries that may be executed repeatedly with different values. I also understood the processing limitations of using a Simple Statement for repeated queries.

## **6. Challenges & Understanding**

The main learning challenge was understanding why repeated execution of a query through a Simple Statement can involve repeated database-side processing. The session material helped clarify the difference by introducing the pre-compilation concept of PreparedStatement.

## **7. Conclusion**

Week 03 strengthened my Java backend fundamentals by introducing JDBC and its statement objects. The session particularly focused on the limitations of Simple Statement and the pre-compiled query concept of PreparedStatement, providing a foundation for database interaction in Java applications.

Source basis: J2SE/JDBC training presentation supplied for Week 03. The report follows the terminology and concepts presented in the supplied material.